Basic Data Sorting & Filtering

Task 11 · Data Analytics Track

Prepared by: Vaishnavi · Veda Technology Internship

Dataset: Retail Sales Dataset (1,000 transactions, 1 Jan 2023 – 1 Jan 2024). Tool: Microsoft Excel (AutoFilter, multi-level Sort, COUNTIFS / SUMIFS / AVERAGEIFS for verification).

1. Objective

Sort and filter a dataset to answer simple business questions and practise basic data exploration.

2. Approach

The Raw_Data tab was kept unchanged (header frozen, filter buttons on). For each of the five questions, filters were combined (several conditions at once, which Excel applies as AND) and a multi-level sort was added. The matching rows are saved on their own sheet (Q1–Q5) and the answer is checked with a live COUNTIFS / SUMIFS / AVERAGEIFS formula on the raw data – every sheet shows "OK – matches".

3. Summary of Answers

#	Answer
1	49 transactions reached the maximum Total Amount of 2,000, together worth 98,000. Electronics has the most (19), then Clothing (17) and Beauty (13).
2	61 Clothing purchases by female customers were worth 500 or more. They total 69,700 (average 1,142.62) – about 44.8% of all Clothing sales.
3	44 Beauty transactions in Q4 2023 had a quantity of 3 or more, covering 149 units and 26,475 in sales (average 601.70).
4	58 Electronics purchases were made by customers aged 18–25, totalling 26,760 (average 461.38) – 17.0% of all Electronics transactions.
5	51 bulk, high-price purchases were made by male customers aged 40+, worth 71,000 in total (average 1,392.16) – 15.6% of all sales. They are spread across Electronics (22), Beauty (16) and Clothing (13).

Question 1

Which transactions reached the maximum Total Amount (2,000), and which category has the most of them?

Filters: Total Amount = 2,000 (the maximum)
Sort: Product Category A->Z, then Date oldest->newest
How to do it in Excel: Data -> Filter. On Total Amount choose Number Filters -> Equals -> 2000. Then Data -> Sort: Product Category (A to Z), add level: Date (Oldest to Newest).

Answer: 49 transactions reached the maximum Total Amount of 2,000, together worth 98,000. Electronics has the most (19), then Clothing (17) and Beauty (13).

First 5 rows of the filtered, sorted result:

Txn ID	Date	Customer	Gender	Age	Category	Qty	Price/Unit	Total Amt
743	2023-01-16	CUST743	Female	34	Beauty	4	500	2000
592	2023-01-24	CUST592	Female	46	Beauty	4	500	2000
577	2023-02-13	CUST577	Male	21	Beauty	4	500	2000
257	2023-02-19	CUST257	Male	19	Beauty	4	500	2000
808	2023-04-01	CUST808	Male	33	Beauty	4	500	2000

Question 2

Which Clothing purchases by Female customers were worth 500 or more, and what did they add up to?

Filters: Product Category = Clothing AND Gender = Female AND Total Amount >= 500
Sort: Total Amount largest->smallest, then Date oldest->newest
How to do it in Excel: Data -> Filter. Product Category: tick only Clothing. Gender: tick only Female. Total Amount: Number Filters -> Greater Than Or Equal To -> 500. Sort by Total Amount (Largest to Smallest), add level Date.

Answer: 61 Clothing purchases by female customers were worth 500 or more. They total 69,700 (average 1,142.62) – about 44.8% of all Clothing sales.

First 5 rows of the filtered, sorted result:

Txn ID	Date	Customer	Gender	Age	Category	Qty	Price/Unit	Total Amt
420	2023-01-23	CUST420	Female	22	Clothing	4	500	2000
561	2023-05-27	CUST561	Female	64	Clothing	4	500	2000
476	2023-08-29	CUST476	Female	27	Clothing	4	500	2000
253	2023-08-31	CUST253	Female	53	Clothing	4	500	2000
626	2023-09-29	CUST626	Female	26	Clothing	4	500	2000

Question 3

Which Beauty transactions in Q4 2023 (Oct–Dec) had a quantity of 3 or more?

Filters: Product Category = Beauty AND Date between 01-Oct-2023 and 31-Dec-2023 AND Quantity >= 3
Sort: Date oldest->newest
How to do it in Excel: Data -> Filter. Product Category: Beauty only. Date: Date Filters -> Between -> 01-10-2023 and 31-12-2023. Quantity: Number Filters -> Greater Than Or Equal To -> 3. Sort by Date (Oldest to Newest).

Answer: 44 Beauty transactions in Q4 2023 had a quantity of 3 or more, covering 149 units and 26,475 in sales (average 601.70).

First 5 rows of the filtered, sorted result:

Txn ID	Date	Customer	Gender	Age	Category	Qty	Price/Unit	Total Amt
51	2023-10-02	CUST051	Male	27	Beauty	3	25	75
276	2023-10-02	CUST276	Female	21	Beauty	4	25	100
524	2023-10-03	CUST524	Male	46	Beauty	4	300	1200
656	2023-10-04	CUST656	Male	29	Beauty	3	30	90
441	2023-10-10	CUST441	Male	57	Beauty	4	300	1200

Question 4

How many Electronics purchases were made by young customers aged 18–25, and how much did they spend?

Filters: Product Category = Electronics AND Age between 18 and 25
Sort: Age youngest->oldest, then Total Amount largest->smallest
How to do it in Excel: Data -> Filter. Product Category: Electronics only. Age: Number Filters -> Between -> 18 and 25. Sort by Age (Smallest to Largest), add level Total Amount (Largest to Smallest).

Answer: 58 Electronics purchases were made by customers aged 18–25, totalling 26,760 (average 461.38) – 17.0% of all Electronics transactions.

First 5 rows of the filtered, sorted result:

Txn ID	Date	Customer	Gender	Age	Category	Qty	Price/Unit	Total Amt
847	2023-04-08	CUST847	Female	18	Electronics	4	300	1200
283	2023-05-08	CUST283	Female	18	Electronics	1	500	500
514	2023-03-01	CUST514	Female	18	Electronics	1	300	300
530	2023-02-05	CUST530	Female	18	Electronics	4	30	120
556	2023-06-04	CUST556	Female	18	Electronics	1	50	50

Question 5

Which high-priced bulk purchases (price 300+ per unit, quantity 3+) were made by Male customers aged 40 and over?

Filters: Gender = Male AND Age >= 40 AND Price per Unit >= 300 AND Quantity >= 3
Sort: Date oldest->newest
How to do it in Excel: Data -> Filter. Gender: Male only. Age: Number Filters -> Greater Than Or Equal To -> 40. Price per Unit: >= 300. Quantity: >= 3. Sort by Date (Oldest to Newest).

Answer: 51 bulk, high-price purchases were made by male customers aged 40+, worth 71,000 in total (average 1,392.16) – 15.6% of all sales. They are spread across Electronics (22), Beauty (16) and Clothing (13).

First 5 rows of the filtered, sorted result:

Txn ID	Date	Customer	Gender	Age	Category	Qty	Price/Unit	Total Amt
180	2023-01-01	CUST180	Male	41	Clothing	3	300	900
522	2023-01-01	CUST522	Male	46	Beauty	3	500	1500
264	2023-01-28	CUST264	Male	47	Clothing	3	300	900
416	2023-02-17	CUST416	Male	53	Electronics	4	500	2000
611	2023-02-24	CUST611	Male	51	Beauty	3	500	1500

4. Interview-Style Notes

Sort vs filter? Sorting only re-orders rows (all data stays visible); filtering hides rows that do not meet the conditions without deleting them.

How do multiple filters work? Filters on different columns combine with AND – a row must meet every condition. Within one column you can tick several values (OR). Clear the filters to see the full data again.

Why keep raw data unchanged? Filtering and sorting in place can hide or reorder rows by accident, so the answers are saved on separate sheets and the raw data stays as received.

Sort order matters: with multi-level sorts the first level is applied first (for example Product Category, then Date).

5. Conclusion

Combining several filters with a clear sort order turns 1,000 raw rows into focused answers – for example, 49 maximum-value orders, 61 high-value Clothing purchases by women, and 51 bulk premium purchases by men aged 40+.